Galaxies Quench When Gas Supply Loses the Race

A step-function response from gas-mass conservation unifies black hole and halo quenching across 350,000 galaxies.

arXiv:2610.20505 — Step-like quenching from gas supply versus depletion timescales

Authors: Zhicheng He, Enci Wang, Luis C. Ho, Yingjie Peng, Huiyuan Wang, Yong Shi , Tao Wang, Yongquan Xue et al.

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Background

Galaxies stop forming stars through many routes, from black hole feedback to halo heating. A single framework linking these diverse quenching mechanisms has remained elusive.

> **No simple, unified physical rule connects the varied quenching mechanisms to observable galaxy populations.**

What this paper is selling

Gas-mass conservation yields a step-function quenching response set by supply versus depletion timescales. Fitting it to observed quenched fractions infers the gas-supply timescale directly, revealing how black holes and halos regulate galaxies.

Key results to remember

- Characteristic gas-supply timescale is about 1 Gyr across the sample.
- Black holes drive inside-out quenching; dark matter halos drive outside-in quenching.
- Intermediate surface-density galaxies sit at a critical state for quenching and rejuvenation.

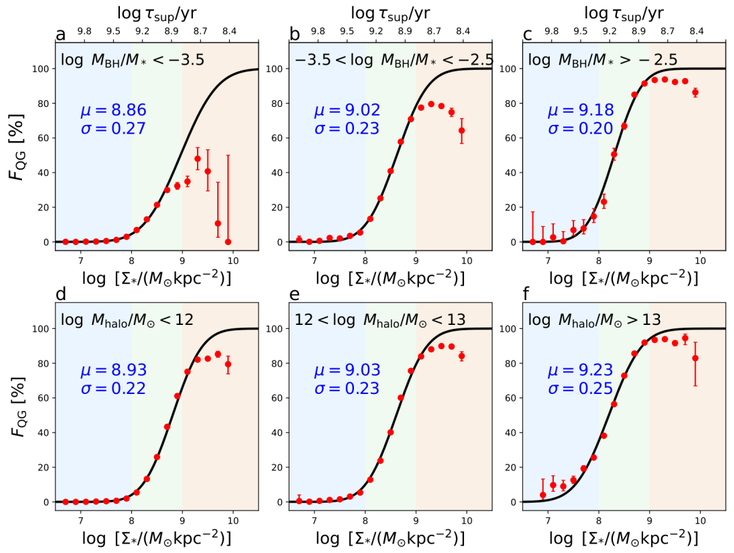


Figure 2 | Black hole growth lengthens and narrows the supply-timescale distribution, while halo mass lengthens and broadens it.

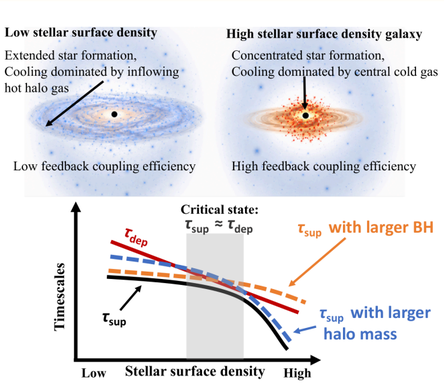


Figure 4 | At intermediate surface density, supply and depletion timescales converge, making quenching most sensitive to regulation.